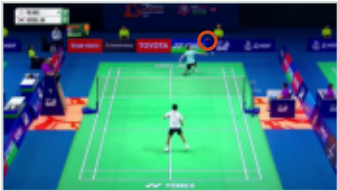


How a height is recovered, and what never sees a pixel

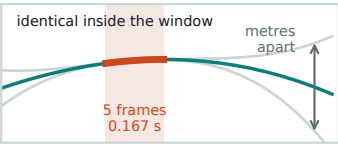
Each panel on the left is the measurement that forced the stage it points to.

a. What the camera gives



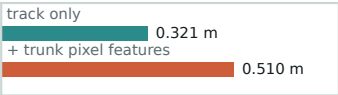
RGB background difference
The shuttle is circled. At 7 px across it is a smudge in RGB and unambiguous in the difference channel - which is why the input has a fourth channel.

b. Why one stroke, not five frames



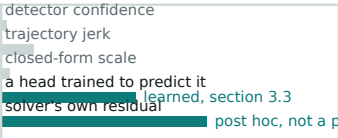
Three flights, one image track. Over 0.167 s the arc is straight and the height is unobservable; a whole stroke is long enough for drag and gravity to bend it.

c. Adding pixels makes it worse



The 64-d trunk feature at the detected point drowns the 14-d track it was meant to help. This is the strongest form of the claim in the title.

d. Who knows when the answer is wrong



Rank correlation with the true 3D error. No signal already present predicts it, but a head trained to does (0.41), which is the route to dropping the solver from the gate.

Broadcast frames

5 consecutive frames, 512 x 288
RGB + background difference

↓ 4 channels x 5 frames

TrackNetV5-Lite trunk (learned)

heatmap head kept and supervised, soft-argmax to (u, v)
F1 95.58 against the released 96.77 - the cost is recall

↓ one (u, v) per frame

One stroke, assembled

the whole flight: 27 frames at the median
plus the venue camera, calibrated once

↓ 27 x (u, v) + camera

Height head (learned)

6-layer transformer, 192-d, over the whole stroke
reads the track and the camera. Never the image.

↓ one height Z per frame

Horizontal position (algebra, not learned)

(X, Y) = xy_from_uvz(G, col3, (u, v), Z)
exact to 0.031 m; regressing it instead costs 0.152 vs 0.113 m

↓ a 3D track, in metres

Refinement, gated (optional)

paper 1's solver, started from OUR estimate, kept only
where its own fit residual is <= 20 px (80% of strokes)

↓ accepted or discarded

3D trajectory in court metres

0.351 m median against paper 1, 6.9% of frames within 5 cm
the trajectory head estimates no camera at inference

☒ learned here ☐ released weights ☒ algebra / optional

Height error reaches 3D error through the closed form with a measured gain of 4.1x: one centimetre of height is four centimetres of position.